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17 September 1970

MILITARY SPECIFICATION
RECEIVING-DECODING GROUP
AN/ARA-63

This Specification has been approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Depart ments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - The equipment, covered by this specification, shall receive and decode approach control guidance signals from the AN/SPN-41 (per SHIPS-T-5368), and AN/TRN-28 (per ELEX-T-18) Transmitting Sets. The system is designed to provide approach guidance to aircraft carriers and air stations. This specification pertains to design requirements of the AN/ARA-63 Receiving-Decoding Group (RDG) only.

*1.2 Classification - The equipment, covered by this specification, shall consist of the following items:

Type Designation		Applicable Paragraph
Radio Receiver	R-1379/ARA-63 R-1379A/ARA-63 R-1379B/ARA-63	3.5.1
Pulse Decoder	KY-651/ARA-63 KY-651A/ARA-63 KY-651B/ARA-63	3.5.2
Receiver Control	C-7949/ARA-63 C-7949A/ARA-63 C-7949B/ARA-63	3.5.3

1.3 Associated equipment - This equipment shall operate with the associated equipment listed in 6.7. However, the associated equipment in 6.7 will not be provided as part of this equipment.

Beneficial comments (recommendations additions, deletions) and any pertinent data which may be use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents - The following documents, of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein. Precedence shall be per 6.3.

SPECIFICATIONS

Military

MIL-W-5088	Wiring, Aircraft, Selection and Installation of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing, Environmental, Airborne Electronic and Associated Equipment
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Panels, Information, Integrally Illuminated
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of
MIL-T-18303	Test Procedures; Preproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment including Ground Support Equipment

STANDARDS

Military

MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of

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MIL-STD-781	Reliability Design Qualifications and Production Acceptance Tests: Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-785	Systems and Equipment Development and Production, Reliability program for
MIL-STD-2074	Failure Classification for Reliability Testing, General Requirements for
MIL-STD-2076	Unit Under Test Compatibility with Automatic Test Equipment, General Requirements for
MIL-STD-2077	Test Program Sets, General Requirements for

PUBLICATIONS

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedure for Selection and Approval of
AR-10	Maintainability of Avionics Equipment and Systems, General Requirements for
EI-559	Avionics Installation Instructions for Receiving-Decoding Group AN/ARA-63

Naval Electronic Systems Command

ELEX-T-18	Transmitting Set, Electronic Guidance Signal, AN/TRN-28(XN-1)(V)

Naval Ship Systems Command

SHIPS-T-5368	Transmitting Set, Aircraft Approach Control with Radome, AN/SPN-41

2.1.1 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications, refer to both the title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement function may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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2) Copies of Naval Electronic Systems Command and Naval Ship System Command Specifications may be obtained upon application of Commander, Naval Electronic Systems Command, Code 0573, Washington, D.C. 20360.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Microelectronic technology shall be used. Microelectronic items shall conform to requirements specified herein.

b. Other parts and materials requirements shall conform to MIL-E-5400.

c. Nonrepairable subassemblies, as outlined in MIL-E-5400 and AR-10, shall be used. Nonrepairable subassemblies shall be reliable in accordance with AR-10.

d. Where previously produced models of this equipment do not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the prior approval of the procuring activity.

3.2.1 Nonstandard parts and materials approval - Approval for the use of nonstandard parts and materials (including electronic tubes, transistors, and diodes), other than microelectronic devices, shall be obtained as outlined in MIL-E-5400. Microelectronic devices shall be approved in accordance with AR-5.

3.2.2 Modular microelectronic sub-assemblies - When used, Modular Microelectronic Sub-assemblies shall meet the requirements of AR-5. Conformal coatings, encapsulants, embedments or potting materials, used with modular assemblies containing integrated circuits and discrete parts, shall be easily removable without damage to the assembly.

3.2.3 Modules - The electronic portions of the equipment shall be modularized in accordance with AR-10.

3.3 Design and construction - The equipment shall conform to all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total weight - The total weight of the equipment, excluding cables, shall be a minimum, consistent with good design, and shall not exceed 15.0 pounds.

3.3.2 Reliability - When required by contract, the contractor shall establish a reliability program per MIL-STD-785.

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3.3.2.1 Operational stability - The equipment shall operate with satisfactory performance, continuously or intermittently, for a period of at least 1000 hours without the necessity for readjustment of any controls that are inaccessible to the operator during normal use.

3.3.2.2 Operating life - The AN/ARA-63 shall have a total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts that require scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in mean time between failure (MTBF) - The production equipment shall meet a specified MTBF of 1000 hours when tested and accepted per 4.4.3 and MIL-STD-785.

3.3.2.4 Time totalizing meter - The equipment shall contain time totalizing meters in accordance with MIL-M-7793 per Table I.

TABLE I. Time totalizing meters.

Unit	Type of Meter
Radio Receiver	Miniature, Digital, 115 volt, 400 Hertz
Pulse Decoder	Miniature, Digital, 115 volt, 400 Hertz

3.3.3 Cabling and connections

3.3.3.1 Cables and connectors - The equipment shall provide for the use of cables and connectors in accordance with MIL-E-5400.

3.3.3.2 Interconnecting cabling - The equipment shall be capable of satisfactory operation using external, interconnecting wiring in accordance with the applicable requirements of MIL-W-5088. The external interconnecting wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements, and provided that the assembly of the cable to its connectors may be easily accomplished. External cables, and that portion of the connectors attached to the cables, shall not be supplied as part of the equipment.

3.3.4 Control panels - All rack or console-mounted control panels shall conform to the applicable requirements of MIL-C-6781. The configuration of all panels shall have procuring activity approval prior to preproduction testing.

3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements of MIL-E-5400.

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3.3.6 Interference control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of MIL-STD-461 and MIL-STD-462, except as modified below:

a. The Limits of CE01 shall be expanded (made less stringent) by 15 db on the 400 Hz power lines.

b. The Narrowband limits of CE03 shall be expanded (made less stringent) by 10 db between 14 KHz and 60 KHz on the 400 Hz power lines. The broadband switching limits shall expanded (made less stringent) by 6 db between 1 MHz and 25 MHz on the 28 volt DC power line.

c. The broadband switching limits of RE02 shall be expanded (made less stringent) by 10 db between 14 KHz and 60 KHz, and 20 db between 3 MHz and 1000 MHz.

3.3.7 Provisions for maintainability - Built-in test features; construction and packaging; provisions for test points; and other maintainability parameters shall be as specified in AR-10 as detailed herein. The built-in test provisions shall be manually activated by a push-button switch on the control panel. The self-test shall be sufficient for pre-flight and post flight checks. It shall be operable in flight and on the ground. BIT indicators shall be mechanically resetable. They shall also be electrically resetable by use of a test set.

3.3.7.1 VAST compatibility - The equipment shall be compatible with the Versatile Avionic Shop Test System (VAST). It shall meet the requirements of MIL-STD-2076 down to the SRA's as defined below. When required by contract, VAST Test Programs shall be furnished in accordance with MIL-STD-2077. If VAST Test Programs exist for the equipment, and changes to the equipment are made which will affect the fault diagnosis procedure, changes to the existing Test Program shall be prepared as part of the equipment changes in accordance with MIL-STD-2077.

Definition of SRA's

a. The following SRA's comprise the Radio Receiver:

(1) Mixer
(2) Local Oscillator
(3) IF amplifer
(4) Built-in test module (BIT)

b. The following SRA's comprise the Pulse Decoder:

(1) Power Supply
(2) Logic Module
(3) Two-card submodule of the Logic Module with the Logic Module removed from the Decoder and separately tested.

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c. The Receiver Control is an SRA.

3.3.8 Nomenclature and nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.3.9 Standard conditions - The conditions of Table II shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

TABLE II. Standard environmental conditions.

Temperature	Room Ambient (+ 25°C ± 5°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	115 + 1.0 VAC rms, 400 Hz, 5 VAC rms, 400 Hz(panel lighting) and + 27.5 ± 0.5 VDC

3.3.10 Service conditions - The equipment shall operate satisfactorily under any of the environmental service conditions, or reasonable combination of these conditions, as specified in MIL-E-5400 for Class 2 equipment, except as modified herein. During + 95°C Intermittent operation, sensitivity shall be at least 66 db below 1 milliwatt.

3.3.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of curve IV of MIL-E-5400, except that the console mounted panel control shall meet curve III.

3.3.10.2 Shock - The console mounted panel control shall be subjected to impact shocks of 5 G's only.

3.3.11 Warm-up time - The time required for the equipment to warm up, prior to operation, shall be kept to a minimum and shall not exceed 1 minute under standard conditions and 5 minutes at extreme service conditions.

3.3.12 Input electrical power

3.3.12.1 Operating power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the power sources of Table III having characteristics and limits as defined in MIL-STD-704. The power required shall not exceed the specified amounts.

TABLE III. Operating power.

a. AC Power (3 phase) 115/200 volts rms, 400 Hz, category B, 60 volt-amperes.

b. DC Power, + 28V, category B, 10 watts

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3.3.12.2 Lighting power - Input power for lighting shall require not more than 0.5 amperes rms at 5 VAC rms.

3.3.12.3 Degraded performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.3.13 Cooling - The equipment shall meet all performance requirements under the conditions of 3.3.10 without the necessity for cooling air.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under all combinations of standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions, will be specified.

3.4.1 Operation - The equipment shall receive and decode pulse-modulated Ku-Band transmission of elevation and azimuth guidance data from the AN/SPN-41 system or the AN/TRN-28 system, and shall provide output signals to drive crosspointers in accordance with aircraft deviations from a fixed approach path.

3.4.2 Accuracy

3.4.2.1 AZ/EL guidance - The accuracy of the crosspointer output signals (from the AN/ARA-63) that provide elevation guidance, shall be within ± 0.10 degree at the reference glide path (3.4.4.1.2) and ± 0.2 degree at deviations of ± 1.4 degree from the glide path. The accuracy of the crosspointer output signals, that provide azimuth guidance, shall be within ± 0.2 degree at the reference course (3.4.4.2.1); ± 0.4 degree at deviation of ± 2.5 degrees from the course; and ± 0.6 degree at deviations of ± 6.0 degrees from the course.

3.4.3 Frequency characteristics

3.4.3.1 Receiver channelization - The equipment shall operate on 20 Ku-Band channels using 10 frequencies. The frequency and intra-pair pulse codes used shall determine the channel of operation. The operating channels are as shown in Table IV.

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TABLE IV. Intra-pair spacing identification azimuth.

Channel	Frequency (± 1 MHz) 1/	Elevation (± 0.2 usec)	Fly-Right (± 0.2 usec)	Fly-left (± 0.2 usec)
1	15.412 GHz	12 usec	14 usec	10 usec
2	15.436	12	14	10
3	15.484	12	14	10
4	15.508	12	14	10
5	15.532	12	14	10
6	15.568	12	14	10
7	15.592	12	14	10
8	15.616	12	14	10
9	15.664	12	14	10
10	15.688	12	14	10
11	15.412	13	15	11
12	15.436	13	15	11
13	15.484	13	15	11
14	15.508	13	15	11
15	15.532	13	15	11
16	15.568	13	15	11
17	15.592	13	15	11
18	15.616	13	15	11
19	15.664	13	15	11
20	15.688	13	15	11

1/ The ± 1 MHz, listed in the table above, is to be used for testing purposes only.

The following frequencies are reserved for distance measuring equipment (DME):

D1 15.460 GHz
D2 15.640 GHz

*3.4.3.2 Adjacent and alternate channel rejection - Operation on the selected Ku-Band channel shall insure maximum sensitivity. Operation on adjacent channels shall cause at least 25 db loss in sensitivity. Operation on any channel at least two removed from the selected channel shall cause a reduction in sensitivity of at least 35 db. This measurement shall be made using an RF pulse with an envelope essentially as shown in Figure 1.

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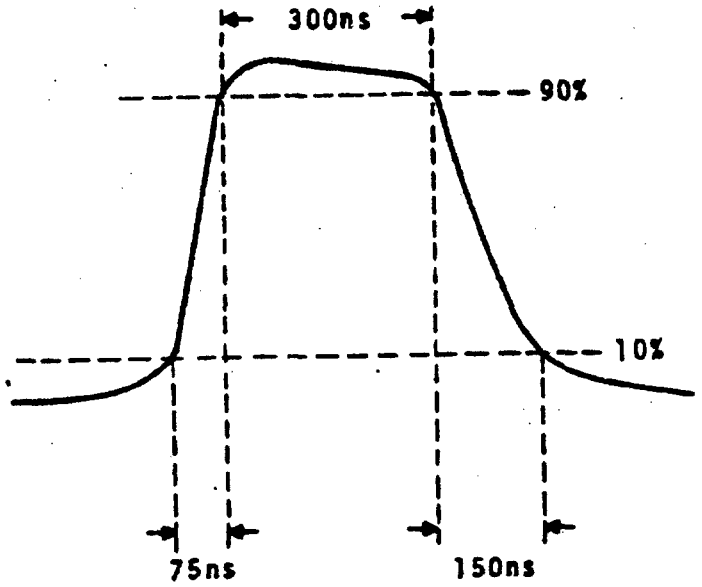


FIGURE 1. Typical transmitted pulse envelope for receiver test purposes.

3.4.3.3 Remote control - The choice of Ku-Band channel shall be made by the pilot by means of a multiposition selector switch located on the Receiver Control.

3.4.4 Input - modulation characteristics - The Ku-Band Radio Receiver and the Pulse Decoder shall accept and correctly interpret signals from the AN/SPN-41 and the AN/TRN-28. Elevation and azimuth guidance signals will be received at the aircraft in an alternating sequence by interception of narrow scanning beams. Each signal comprises a brief series of grouped pulses, having an amplitude envelope corresponding to the radiation pattern of the beam. Such signals are received intermittently, with relatively long and essentially constant intervals between successive interceptions of the same beam.

3.4.4.1 Elevation guidance signals - Elevation guidance signals are transmitted from the AN/SPN-41 or AN/TRN-28 system a minimum of 5 to a maximum of 8 times per second and are obtained by interception of a beam. The received 3 db beamwidth will be 1.3 degrees or less in elevation. The received signal will consist of a pulse train of variably spaced, 0.3 microsecond pulses transmitted in pairs. The signal conveys the desired intelligence by means of the spacing between the pulse pairs. The interval is measured from leading edge to leading edge of the second pulse in each pulse pair. The interval is translated into angle data by the decoder. Channel identification is given by means of the intra-pair spacing within each pulse pair, measured from leading edge to

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leading edge. For elevation signals, Channels 1 through 10 are allotted a 12 microsecond spacing and channels 11 through 20 are allotted a 13 microsecond spacing. Elevation angle data are received from 20 degrees above the horizon to the horizon in a sector whose origin is the landing zone.

3.4.4.1.1 Elevation deviation - Angular deviations of the central ray of the elevation scanning beam from the reference glide path, as the beam scans 5 or more times per second, is defined by inter-pair spacing. The deviation from the reference glide path is 2 Phi microseconds per degree inter-pair spacing where Phi is the positive or negative deviation in degrees. The transmitted inter-pair spacings are accurate within ± 0.1 microsecond of values so calculated; except that a fixed spacing of 60 microseconds is substituted for all values less than 60 microseconds. No transmissions are made at angles below the horizon nor more than 20 degrees above the horizon.

3.4.4.1.2 Reference glide path coding - A reference glide path is defined by an inter-pair spacing of 66 microseconds. This glide path is established at the transmitter and is adjustable in $1/4^{0}$ increments between 2.5 and 5.0 degrees above the horizon. It shall also be possible to obtain other glide paths by a maintenance change in the Pulse Decoder. These glide paths shall be between 2.0 and 5.0 degrees above the horizon in half-degree increments.

3.4.4.2 Azimuth guidance signal - Azimuth Guidance signals are transmitted from the AN/SPN-41 or AN/TRN-28 system a minimum of 5 to a maximum of 8 times per second and are obtained by interception of a beam. The received 3 db beamwidth will be 2.0 degrees or less in azimuth. The received signal will consist of a pulse train as described in 3.4.4.1. For Azimuth signals in the fly-left region (to the right of centerline as viewed from an approaching aircraft), channels 1 through 10 are allotted a 10 microsecond intra-pair spacing and channels 11 through 20 are allotted an 11 microsecond intra-pair spacing. For Azimuth signals in the fly-right region, channels 1 through 10 are allotted a 14 microsecond intra-pair spacing and channels 11 through 20 a 15 microsecond intra-pair spacing. The transition from 10 to 14 microseconds and from 11 to 15 microseconds occurs at the instant the center of the scanning beam crosses the approach centerline. Azimuth angle values are encoded and transmitted throughout a 40 degree sector (± 20^{0} about the centerline).

3.4.4.2.1 Reference course coding - A reference course is defined by an inter-pair spacing of 60 microseconds. This code spacing is transmitted when the central ray of the azimuth scanning beam is directed along the desired approach centerline (normally runway or carrier flight deck centerline).

3.4.4.2.2 Azimuth deviation coding - Angular deviations, of the central ray of the azimuth scanning beam from the reference course, are defined by inter-pair spacings equal to 60 + 2 Theta microseconds, where Theta is either left or right deviation in degrees. The transmitted inter-pair spacing are accurate within ± 0.20 microsecond of values calculated from this relation. The left-right sense of any deviation is represented by intra-pair spacings as specified in 3.4.4.2. Illustrative azimuth code spacings are listed in Table V.

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TABLE V. Azimuth angle coding examples identity code pair spacings.

Course Deviation Region	2/ Angle-Degrees	Intra-Pair Spacing Channels 1-10 Microseconds	Intra-Pair Spacing Channels 11-20 Microseconds	Inter-Pair Spacing Microseconds
	+ 20	10	11	100
Fly-left	+ 10	10	11	80
	+ 3	10	11	66
	+ 2	10	11	64
	0	10 or 14	11 or 15	60
	- 2	14	15	64
Fly-right	- 3	14	15	66
	- 10	14	15	80
	- 20	14	15	100

2/ Fly-left deviations denoted by "+"; fly-right by "-"

3.4.4.3 Station identification - The ground subsystem transmits station identification information by variation of the intra-pair pulse spacing of the azimuth transmissions. The station identification message consists of Morse code characters (normally three). The station identity message is transmitted only on the azimuth beam. It is contained within a group of 15 pulse pairs, the first four of which have an intra-pair spacing slightly larger than the nominal. A mark (tone ON) is encoded by increasing the intra-pair pulse spacing by 0.5 microsecond. A space (tone OFF) is represented by the nominal intra-pair pulse spacing. The combination of marks and spaces, forming a Morse code character, is transmitted sequentially, one per scan. The remainder of the 15 pairs have normal or increased intervals in accordance with a particular binary code. The AN/ARA-63 video output shall preserve this information, but decoding shall be accomplished in equipment external to the AN/ARA-63.

3.4.4.4. Obstacle clearance - When desired, the elevation transmitter may indicate an angle below which the aircraft would encounter an obstacle during its approach or landing maneuver. This indication is given by increasing the intra-pair interval of all pulse pairs transmitted while the antenna is inclined below the selected angle. For all channels, the incremental spacing is 0.5 microseconds. The AN/ARA-63 video output shall preserve this information, but decoding shall be accomplished in equipment external to the AN/ARA-63.

3.4.4.5 Auxiliary data - The Auxiliary data requirements of this specification shall be processed in equipment external to the AN/ARA-63. When desired, the obstacle warning signal will be encoded with the elevation guidance signal. Station identification will be encoded with the azimuth guidance signal.

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3.4.5 Dynamic range and coverage

3.4.5.1 Signal amplitude response - The Ku-Band Radio Receiver shall utilize circuits such that the system response to the coded information shall be essentially the same when the RF input power is within the range of 72 db below 1 milliwatt and 10 db below 1 milliwatt. As the receiver sensitivity varies from maximum to the value obtained at a distance of 1/2 nm from the transmitter, the system shall utilize guidance information between 0 db and - 20 db on the envelope of each main beam signal. Within the specified system accuracy, separate elevation and azimuth memories shall be effective in establishing the respective levels necessary to determine aircraft position relative to the required flight path.

3.4.5.2 Angular coverage - The Pulse Decoder shall produce proportional (analog, "slowly varying DC voltage") output signals that represent angular deviations from the reference glide path nominally between 0 and ± 1.4 degrees and angular deviations from the reference azimuth course nominally between 0 and ± 6.0 degrees. Within these angular ranges, the currents supplied to the crosspointer circuits shall represent the average angle values encoded in the received signals to within the accuracies specified in 3.4.2, and shall vary smoothly among intermediate values. Adjustment of meter drive current, to vary course sensitivity, shall be provided as a maintenance change in the Pulse Decoder.

3.4.6 Sensitivity - Angle decoding sensitivity at 6°, using a terminated RF signal generator, shall be at least 72 db below 1 milliwatt. Angle decoding sensitivity, at other decoding angles, shall be at least 70 db below 1 milliwatt. When angle decoding is enabled, the flags shall be driven from view at the 72 db below 1 milliwatt sensitivity.

3.4.7 Crosspointer indicator and flag operation - The Pulse Decoder shall be capable of driving the vertical and horizontal deviation needles and the flag coils of the meter movements per 6.7.2.

3.4.7.1 Elevation display - When the received elevation RF signal strength is adequate for tracking, the flag associated with the horizontal needle shall be deflected out of view and the horizontal needle shall be controlled by the received decoded angle.

a. The voltage supplied to deflect the horizontal flag out of view shall be 500 millivolts ± 100 millivolts.

b. When the input signal coding is set to the reference glidepath angle, the horizontal needle voltage shall be 0 volt ± 100 mv.

c. The voltage shall increase proportionately until the input signal coding equals + 1.4^{o} deviation from the reference glidepath. At ± 1.4^{o}, the voltage shall be ± 2.20 Volts ± 0.20 volt. At + 17° and - 3° deviation from the reference glidepath, the voltage shall be - 3.3 volts ± 0.3 volt and + 3.3 volts ± 0.3 volt, respectively.

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d. For input signal coding outside the limits of (c), the voltage shall continue to increase. For angles in excess of ± 1.4°, this increase shall be non-linear. At the scan limits, the voltage shall be ± 3.3 volts ± 0.3 volt.

e. In the absence of received RF signals, the horizontal needle shall be deflected to the top of the meter by a drive of at least 7.0 volts to indicate a fly-up condition. The corresponding flag voltage level shall drop to 0 volt ± 0.1 volt (and the flag will appear).

*3.4.7.2 Azimuth display - When the received azimuth RF signal strength is adequate for tracking, the flag associated with the vertical needle shall be deflected out of view and the vertical needle shall be controlled by the received decoded angle.

a. The voltage supplied to deflect the vertical flag out of view shall be 500 millivolts ± 100 millivolts.

b. When the input signal coding is set to the reference azimuth course, the vertical needle voltage shall be 0 volt ± 50 mv.

c. The voltage shall increase proportionately until the input signal coding equals ± 6.0° deviation from the reference azimuth course. At ± 6.0°, the voltage shall be ± 2.20 volts ± 0.20 volt.

d. For input signal coding outside the limits of (c), the voltage shall continue to increase. For angles in excess of ± 6°, this increase shall be non-linear. At ± 20° deviation from the reference course, the voltage shall be ± 2.6 volts ± 0.3 volt.

e. In the absence of received RF signals, the vertical needle shall be deflected to the right of the meter by a drive of at least 7.0 volts to indicate a fly-right condition. The corresponding flag voltage level shall drop to 0 volt ± 0.1 volt (and the flag will appear).

3.4.7.3 Flag operation with radio receiver not energized - In the absence of main power to the Radio Receiver, the flag voltages shall be 0 volt ± 0.10 volt (the flags will appear) and the vertical and horizontal needle output voltages shall be 0 volt ± 0.20 volt (the vertical and horizontal needles will be at the center of the indicator).

3.4.8 Distance measuring equipment (DME) - When DME information is transmitted as specified in ELEX-T-18, the Radio Receiver shall be capable of providing the detected video signals to a separate DME equipment external to the AN/ARA-63 for decoding.

3.4.9 Station identification - The Pulse Decoder shall be capable of accepting the variations in beam coding specified in 3.4.4.3 for transmitting Station Identification while continuing to recognize the intended direction message. For example, if the received signal contains intra-pair spacing of 10.5 microseconds instead of 10 microseconds, the decoder shall continue to interpret the message as an azimuth transmission with a fly-left instruction. The Pulse Decoder shall also have the capability of providing the identity message to an auxilliary item of equipment for decoding and display whenever such an item is available in the aircraft.

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3.4.10 Obstacle clearance - The Pulse Decoder shall be capable of accepting the variations in beam coding specified in 3.4.4.4 for transmitting obstacle clearance information while continuing to recognize the intended direction message. It shall also have the capability of providing the obstacle clearance message to an auxilliary item of equipment for decoding and display whenever such an item is available in the aircraft.

*3.4.11 Sidelobe cancellation - The equipment shall recognize and react, in an operationally acceptable manner, to information received from sidelobes. The Ku-Band Radio Receiver shall not provide tracking for sidelobe signals less than - 40 dbm.

3.4.12 System built-in-test (BIT) - System BIT shall be included in the design. It shall check the Angle guidance functions on an end-to-end basis. Normal operation shall be presented to the pilot as known AZ and EL deviations with flags pulled. The system BIT mode shall be activated with a single control.

3.5 Detail requirements - The installation details of the AN/ARA-63 shall be in accordance with EI-559.

3.5.1 Radio receiver - The Radio Receiver shall be per 1.2 and shall meet the following requirements.

3.5.1.1 Function - The Radio Receiver shall transform the incoming pulse modulated Ku-Band transmissions to video pulses suitable for processing in the Pulse Decoder.

3.5.1.2 Form factor - The Radio Receiver shall be no longer than 5-1/4" wide by 2-1/2" high by 8-1/2" deep, excluding connectors.

3.5.1.3 Weight - The weight of the Radio Receiver shall not exceed 5 pounds.

*3.5.1.4 Contents - The Radio Receiver shall contain the following:

a. Front End Assembly - The front end assembly shall consist of:

(1) Local Oscillator - The local oscillator shall be a crystal-controlled, solid state unit employing multipliers, amplifiers and filters to obtain the required output.
(2) Microwave Mixer - The mixer shall be a balanced, waveguide mixer.
(3) IF Amplifier - The IF amplifier unit shall have a center frequency of 150 MHz and bandwidth of 10 MHz. In addition, the unit shall contain a detector and video amplifier.

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(4) Waveguide shutter switch - The R-1379A/ARA-63 and R-1379B/ARA-63 Radio Receivers shall have a waveguide shutter switch assembly to protect the mixer crystals when the equipment is OFF.

b. Receiver BIT Module - The Receiver BIT module shall contain the following:

(1) An IF oscillator for test signal injection into the Receiver's IF amplifier.
(2) A mixer crystal current monitoring circuit.
(3) A video activity monitoring circuit for checking the Receiver's output.
(4) Provision for operating the Receiver's BIT failure indicator, based on outputs from the monitoring circuits, (2) and (3).
(5) The R-1379B/ARA-63 (only) Radio Receiver BIT input circuit shall have a matching network that provides proper impedance matching for the BIT modulation. This characteristic impedance shall be 75 ohms.

3.5.1.5 Electrical connections - Connections to external circuits shall be provided per Table VI.

TABLE VI. Radio receiver electrical connectors

Reference Designation	Receptacle Type	Function
1J1	M81511/01E-14-37P1	Power-BIT-Channel Select
1J2	RF Coax connector, OSM3/ 610-6205 or equivalent	Video Output
1J3	RF Coax connector, OSM3/ 610-6205 or equivalent	Test Input
1P1	UG/1665/U (Waveguide)	Antenna Input

3/ Omni Spectra Minature

3.5.2 Pulse decoder - The Pulse Decoder shall be per 1.2 and shall meet the following requirements.

3.5.2.1 Function - The Pulse Decoder shall process the incoming video pulses and provide command and flag signals in both azimuth and elevation. These signals will be used to drive the crosspointer indicator in the cockpit.

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3.5.2.2 Form factor - The Pulse Decoder shall be no larger than 4-1/4" wide by 4-1/2" high by 11" deep, excluding connectors.

3.5.2.3 Weight - The weight of the Pulse Decoder shall not exceed 9 pounds.

* 3.5.2.4 Contents - The Pulse Decoder shall contain the following:

a. Power Supplies - Direct-current Power Supplies shall be per * Table VII.

TABLE VII. Pulse decoder DC power supplies.

Nominal Voltage (Volts DC)	Voltage Tolerance (Percent)	Ripple Less Than (Millivolts)
+ 20	± 5	25
+ 10	± 5	25
+ 5	± 5	50
- 10	± 5	25

b. Clock/flag module - The clock/flag module shall:

(1) Supply the clock frequencies for timing the decoder.
(2) Check the video signals for proper tracking information and provide for flag activity if the information is erroneous.
(3) Provide video coding signals for modulating the IF oscillator in the Receiver BIT module. The KY/651B/ARA-63 (only) Pulse Decoder BIT output circuit shall be capable of driving an impedance of 75 ohms in parallel with a 3000 pf capacitor.

c. Video-identity module - The video-identity module shall receive video from the IF amplifier and perform a multi-level threshold detecting process. It shall also identify the intra-pair pulse spacing being received as to azimuth or elevation information.

d. Error module - The error module shall receive the video from the video-identity module and compare the spacing of the angular data pulses with reference angle timing pulses. It shall produce an output signal that is proportional (in pulse width) to the angular error of the aircraft from the glide path.

e. AGC module - The AGC module shall develop a DC voltage that shall control the Receiver IF amplifier gain. This DC control voltage shall:

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(1) Be logarithmically proportional to the RF input to the Receiver (db input/volts AGC).

(2) Provide for Receiver (IF) desensitization if antenna sidelobe and/or multi-path reflections are recognized

f. Memory module - The memory module shall convert the output of the error module into a DC voltage that is proportional to the error. This DC output voltage shall be proportional to the average angular error of the aircraft. The memory module shall contain the Decoder BIT module that shall monitor the DC azimuth and elevation voltages produced when the BIT button of the Control Receiver is depressed. If these voltages are significantly incorrect the Decoder BIT failure indicator shall operate.

g. The KY-651A/ARA-63 and KY-651B/ARA-63 shall contain a glide slope selector switch for ease in glideslope adjustment.

3.5.2.5 Electrical connections - Connections to external circuits shall be provided per Table VIII.

*TABLE VIII. Pulse decoder electrical connectors.

Reference Designation	Receptacle Type	Function
2J1	~~MS1511/01E-14-37P3~~	~~Power Input, Deviation and Flag Outputs~~
2J2	MS1511/01E-14-37S1	Power Output, Control
2J3	MS1511/01E-14-37S2	Test Connector
2J4	RF Coax connector OSM5/ 610-6205 or equivalent	Test Connector Test IF to Receiver
2J5	RF Coax connector OSM5/ 610-6205 or equivalent	Test Connector Test IF Input
2J6	RF Coax connector OSM5/ 610-6205 or equivalent	Video input from Radio Receiver

5/ Omni Spectra Miniature

3.5.3 Receiver control - The Receiver Control shall be per 1.2 and shall meet the following requirements.

3.5.3.1 Function - The Receiver Control is used to turn power ON and OFF, to select the operating channel, and to initiate system BIT test.

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3.5.3.2 Form factor - The Receiver Control shall be no larger than 5-13/16" wide by 1-9/16" high by 3-11/16 deep, not including protrusion of controls (per 3.5.3.4) and the connector.

3.5.3.3 Weight - The weight of the Receiver Control shall not exceed 1 pound.

3.5.3.4 Controls - The Receiver Control shall provide controls per Table IX.

TABLE IX. Receiver control - controls.

Name	Type	Function
ON-OFF	Toggle (Switch)	Controls power to AN/ARA-63
CHANNEL	one rotary (discrete switch)	Selects one of twenty operating channels in the Radio Receiver
BIT	Push-to-Test Button (Switch)	Manually initiates system BIT tests in the AN/ARA-63

3.5.3.5 Electrical connections - Connections to external circuits shall be provided per Table X.

TABLE X. Receiver control electrical connector.

Reference Designation	Receptacle Type	Function
3J1	M81511/01E-14-37P1	Power, BIT, Channel Select

*3.5.3.6 Panel lighting - Panel lighting shall be as specified in MIL-P-7788. The C-7949B/ARA-63 (only) Receiver Control panel information markings shall have uniform illumination and preclude the need for additional indicator lights for function identification, e.g., it shall not have a power ON/OFF lamp.

3.5.4 BIT function - The AN/ARA-63 shall operate with any of the crosspointer meters listed in 6.7.2. The meter will not be provided as part of the AN/ARA-63 subsystem. Azimuth guidance information is presented to the vertical needle on the meter, and elevation guidance to the horizontal needle. Depression of the BIT push-button on the Receiver Control shall cause:

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a. The crosspointer needles to take the following positions:
Vertical needle - slowly oscillating 1/2 scale fly-left/ fly-right
Horizontal needle - on glide slope

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance on the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of tests - Equipment covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

a. Preproduction (First Article) tests
b. Initial production tests
c. Acceptance Tests
d. Life Test

4.2 Preproduction (first article) tests - Preproduction tests shall be made on a set of equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests, when so desired. A contractor, not having adequate facilities to conduct required tests, shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (first article) test data - The contractor shall submit all data, collected in conducting these tests, to the acquisition activity for review.

4.2.2 Scope of tests - Preproduction tests shall include all tests deemed necessary by the acquisition activity, to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of MIL-T-5422.

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4.2.3 Preproduction (first article) approval - Approval of the preproduction sample shall be by the acquisition activity upon satisfactory completion of all tests. No production equipment shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment, prior to approval of the preproduction sample, is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the production equipments deliverable under the contract.

4.2.4 Production equipments - Equipments supplied under the contract shall, in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and, for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the acquisition activity.

4.3 Initial production tests - One of the first ten production equipments shall be selected and sent, at the contractor's expense, to a designated Government laboratory for tests. This equipment shall be selected by the acquisition activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of tests - The equipment may be subjected to any and all tests that the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance, and quality and that it meets all applicable requirements.

4.3.2 Accessory material - In addition to the complete equipment submitted for initial production tests, the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial production sample approval - Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material, or performance defect, made evident during these tests, shall be corrected by the contractor to the satisfaction of the acquisition activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment, under the contract, to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted.

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4.3.4 Reconditioning of initial production test sample - On completion of the initial production test, the equipment shall be reworked by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance tests - All inspection and testing shall be under the supervision of the Government inspector. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

a. Individual tests
b. Sampling tests
c. Reliability assurance tests
d. Special tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

a. Examination of product
b. Operational test
c. Manufacturing run-in test

4.4.1.1 Examination of product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing run-in test - Each equipment shall be operated under the conditions specified in Table XI for a period of 50 hours without failure. A failure shall be defined as anything that causes malfunctioning of the equipment. Only those adjustments will be permitted that can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment. This test shall be deleted if the reliability test includes a test of each equipment consuming at least 50 hours of operation.

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TABLE XI. Manufacturing run-in test environmental conditions.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of ± 3 G's.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of a 50-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. The direction of vibration shall be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions, the vertical direction shall be used. During the 50-hour period of operation, following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it shall be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain that the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment and previous equipment. A record of all failures shall be kept. The 50 hour period specified above may be composed of several shorter periods to conform with standard working hours.

4.4.2 Sampling tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with the following (sampling tests shall be conducted only when reliability assurance tests of 4.4.3 are deleted by contractual action):

Quantity of Equipments Offered for acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
Additional	1 for each additional 200 or fraction thereof

One out of the first ten need not be selected and tested if initial producton tests are conducted. Sampling tests are not required when Reliability Assurance Tests are conducted.

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4.4.2.1 Scope of tests - as a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.
b. Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.
c. Manufacturing run-in test specified in 4.4.1.3, except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability assurance tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests, as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase of MIL-STD-781, shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and MIL-STD-2074.

4.4.3.1 Qualification phase - Prior to the acceptance of equipment under the contract or order, a minimum of three AN/ARA-63 sets shall be tested in accordance with MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipment sets to be tested shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level F shall be used. The Accept-Reject criteria for Test Plan III shall be used. Classification of failures shall be in accordance with MIL-STD-781 and MIL-STD-2074.

4.4.3.2 Reliability production assurance (sampling) tests - The production equipment shall be tested in accordance with MIL-STD-781 (as modified herein) under section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests". Test Level F shall be used. Classification of failures shall be in accordance with MIL-STD-781 and MIL-STD-2074.

*4.4.3.2.1 All equipment test - Each equipment produced shall be tested for 120 hours actual ON time. Prior to the actual-ON-time-120-hour test for each equipment, an additional burn-in period may be used at the option of the contractor. If the burn-in period is to be used, the details thereof must be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract, the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time)

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shall be totaled and the results compared with the reject line of test, Plan III of MIL-STD-781. (Extend the line as necessary to accomodate the data). These totals shall accumulate so that, for any one time, the experience from the beginning of the contract is included. At the conclusion of each month, the test results shall be sent to the acquistion activity and to the Naval Air Systems Command, to the attention of the specification cognizant code per 6.8. At any time that the current totals of test hours and test failures plotted on Test Plan III curves show a reject situation, the acquistion activity shall be notified. The acquisition activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists, pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

4.4.3.3 Test details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the acquisition activity prior to the beginning of the qualification test phase of the Reliability Assurance Tests.

4.4.4 Special tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment, selected for special tests, may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special test schedule - Selection of equipments for special tests shall be made as follows:

a. On an early equipment, after an engineering or material change.
b. Whenever failure reports or other information indicate additional tests are required. (This will be determined by the acquisition activity).

4.4.4.2 Scope of tests - Special tests shall consist of such tests as approved by the acquisition activity. Test procedures previously approved for the preproduction tests shall be used, where applicable. Where not applicable, the contractor shall prepare a test procedure and submit it to the acquisition activity for approval, prior to conducting the tests.

4.4.5 Equipment failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

a. Determine the cause of failure.
b. Determine if the failure is an isolated case or design defect.

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c. Proposed corrective action (intended to reduce the possibility of the same failure(s) occuring in future tests) shall be submitted to the acquisition activity for approval.

d. Where practical, include a test in the individual tests (to check all equipment for the reoccurrence of the above defect) until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual tests. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the Following. (Equipments that have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests). When reliability tests are conducted, the life test shall be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
Additional	1 for each additional 500 or fraction thereof

4.5.1 Test conditions - The life test shall be conducted under the simulated service conditions per Table XII.

TABLE XII. Life test environmental conditions.

Temperature	Normal room
Altitude	Normal ground (0-5000 ft.)
Humidity	Room ambient
AC Voltage	115 + 5 volts rms (at lowest applicable frequency) 5 VAC rms (panel lighting)
DC Voltage	+ 27.5 ± 2.0 volts

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4.5.2 Test periods - The test may be run continuously or intermittently. Each period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned ON and OFF several times and put through its various phases of operation.

4.5.3 Performance check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subjected to approval by the acquisition activity.

4.5.4 Test data - The contractor shall keep the daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of a part failure, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

a. Part type number
b. The circuit reference symbol number
c. The part function
d. Name of manufacturer
e. Nature of the failure
f. The number of hours that the part operated prior to failure.

4.5.4.1 Failure report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the acquisition activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures that occurred. The acquisition activity will review such proposals and determine whether they are acceptable.

4.6 Test procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the acquisition activity for review and approval. The right is reserved by the acquisition activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the acquisition activity. However, the right is reserved by the acquisition activity to require modification of such procedures, including additional tests, when deemed necessary.

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4.7 Reconditioning of tested equipment - Equipment that has been subjected to initial production acceptance and life tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking, the contractor shall perform the Acceptance individual operational test (per 4.4.1 b) and then resubmit the equipment for acceptance.

4.8 Presubmission testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and retest - Equipment that has been rejected may be reworked or have parts replaced to correct the defects and be resubmitted for acceptance. Before resubmitting, full particulars, concerning previous rejection and the action taken to correct the defects found in the original, shall be furnished the Government inspector.

5. PACKAGING

All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794. In the event that the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6.0 NOTES

6.1 Intended use - The Receiving - Decoding Group (RDG) AN/ARA-63 (Airborne Microwave Landing Subsystem) is used in conjunction with the surface based Transmitting Sets AN/SPN-41 or AN/TRN-28 to provide either primary or backup instrument approach capability for aircraft. The transmitting set provides guidance information for any number of aircraft at a range to 20 miles. The information is used by the pilot to maintain the aircraft on the approach path toward the touchdown point.

The transmitting set consists of two scanning beam antennas, their associated pulse transmitters and control equipment. At a range of 20 miles, the azimuth scanning beam sweeps an area approximately 8 miles wide, while the elevation or glide path beam is approximately 4 miles high.

The coded, angular information, from the ship or shore installation, is received in the aircraft Radio Receiver, decoded by the Pulse Decoder, and displayed on a crosspointer meter. The vertical needle of the crosspointer is used for azimuth guidance and the horizontal needle for vertical guidance when an aircraft is approaching the carrier or runway. With both needles centered,

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the aircraft is at the deck or runway centerline and the reference glide path. In the event of a failure or inconsistent data, a flag will be presented to the pilot in conjunction with a fly-up or a fly-right needle deflection.

6.2 Ordering data - Procurers may exercise any desired option(s) offered herein. Procurement Documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing.

6.3 Precedence of documents - When the requirements of the contract, this specificaton or applicable subsidiary specification are in conflict, the following precedence shall apply:

a. Contract - The contract shall have precedence over any specification.
b. This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications, where applicable, shall be specifically prior approved in writing by the acquisition activity.
c. Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance objectives - Simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions, beyond the requirements of this specification, are objectives that shall be considered in the production of this equipment. Where it appears that a substantial reduction in weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes, other than those specified in MIL-E-5400, it is desired that their use be investigated. When investigation shows that advantages can be realized, a request for approval shall be submitted to the acquisition activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Type designations - The equipment type designation may be modified by the acquisition activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The correct type number shall be used on nameplates, shipping records and instruction books, as applicable.

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6.6 Revisions - In specification revisions and superseding amendments, an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.7 Associated equipment

6.7.1 Test set, decoder-receiver group AN/ARM-146

6.7.1.1 Function - The Test Set may be used to supply simulated azimuth and elevation signals to the Radio Receiver at the intermediate frequency (IF). The Test Set permits selection of various approach angles. It measures the deviation of the output signals, thus providing a measurement of Pulse Decoder accuracy. The power level of the IF source may be controlled by a (25 db) one step built-in attenuator.

6.7.2 Crosspointer Meter - The AN/ARA-63 azimuth and elevation signals will be properly displayed by one of the following: ID-249, ID-351, ID-811, ID-1329, ID-1144 or AVA-1.

6.7.3 Radio receiver antenna - The Ku-Band antenna(for angle data), to be used in conjunction with the Radio Receiver, will be vertically polarized. The coverage will be + 10^{0} to - 20^{0} in elevation and $\pm$ 35^{0} in azimuth. At least 4 db of gain will be available throughout the coverage. Peak-to-peak ripple in the coverage will be less than 3db and the SWR into a connecting RG-91/U waveguide will be less than 1.5:1.

6.8 Standard measurements - Throughout this specification, the following shall apply:

a. Each mileage distance shall be slant range distance from the aircraft to the point-of-touch-down, unless otherwise specifically defined.
b. Each mileage distance shall be nautical miles (nm).
c. Nautical miles shall be 6076.115 feet (per the international standard, used officially in the U.S. since 01 July 1959).
d. Knots shall be nautical miles per hour, based upon (c) above.

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6.9 Metric conversion - Throughout this specification, the following measurements shall have the listed metric equivalents.

Measurements	Metric Equivalent
1 - 9/16 in.	3.96875 cm.
2 - 1/2 in.	6.35 cm.
3 - 11/16 in.	9.36625 cm.
4 - 1/4 in.	10.795 cm.
4 - 1/2 in.	11.43 cm.
5 - 1/4 in.	13.335 cm.
5 - 13/16 in.	14.76375 cm.
8 - 1/2 in.	21.59 cm.
11 - in.	27.94 cm.
4 - mi.	6.4 Km.
8 - mi.	12.8 km.
20 - mi.	32.0 km.
1 - lb.	0.45 Kg.
5 - lbs.	2.25 kg.
9 -lbs	4.05 kg.
15 - lbs.	6.75 kg.

6.10 Data requirements - Deliverable data required for this specification is cited in the following paragraphs:

Paragraph	Data Requirements	Applicable DID
4.2.1	Preproduction (First Article) Test Data	UDI-I-21349
4.5.4.1	Failure Reports	DI-R-2118
4.6	Test Procedures	UDI-T-21347

6.11 Specification cognizance - This specification is under cognizance at NAVAIR-5334.

Preparing Activity
Navy-AS
Project No. 5841 - N006

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MIL-R-81498C(AS) Receiving-Decoding Group AN/ARA-63

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

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